

Quality Control for Assessing the Reliability of Radiometric Measurements at the Luisha Mining Site, Haut-Katanga Province, Democratic Republic of the Congo

Jacques Nakamwambila Kiadimuyika¹; Evariste Kazadi Tshiamala¹;
Eli-Achille Manwana Mfumukani³; Pierre K. Kaseti²

¹Department of Space Physics of Remote Sensing, Doctoral School of Telecommunication and Space Remote Sensing (ETS), National Pedagogical University (UPN), Kinshasa, Democratic Republic of the Congo.

²Department of Geology, Faculty of Sciences and Technologies, University of Lubumbashi (UNILU), Lubumbashi, Democratic Republic of the Congo.

³Department of Department of Gravimetry and Geodesy, Center of Research in Geophysics (CRG), Kinshasa, Democratic Republic of the Congo.

Publication Date: 2026/09/18

Abstract: This paper evaluates the integrity of radiometric observations collected across the Luisha mining area before those data are used for environmental interpretation or rehabilitation planning. Natural terrestrial radiation was measured along ten field profiles using a Geiger counter. Quality assessment combined descriptive statistics, geostatistical modelling and frequency-domain processing. Kriging was used to characterize spatial prediction uncertainty and to separate the survey area into zones of differing confidence according to station distribution. The histogram and normal Q-Q plot show an approximately Gaussian central population, although the upper tail remains positively skewed because of a small number of high readings. A low-pass FFT filter, implemented with a two-point window and a 0.25 Hz cutoff frequency, further improved the signal-to-noise ratio. After these checks and corrections, the dataset is considered suitable for subsequent radiometric mapping, interpretation and modelling.

Keywords: Natural Terrestrial Radioactivity; Radiometric Measurements; Quality Control; Geostatistics; Kriging; FFT.

How to Cite: Jacques Nakamwambila Kiadimuyika; Evariste Kazadi Tshiamala; Eli-Achille Manwana Mfumukani; Pierre K. Kaseti (2026). Quality Control for Assessing the Reliability of Radiometric Measurements at the Luisha Mining Site, Haut-Katanga Province, Democratic Republic of the Congo. *International Journal of Innovative Science and Research Technology*, 11(8), 3857-3864. <https://doi.org/10.38124/ijisrt/26aug1445>

I. INTRODUCTION

Luisha is a mining area in Kambove Territory, Haut-Katanga Province, roughly 80 km northwest of Lubumbashi. Geologically, it belongs to the Greater Katanga Copperbelt, one of the world's major metallogenic provinces for copper and cobalt and a region that also hosts tin, precious-metal and iron occurrences. The exceptional economic importance of these deposits, particularly for Cu and Co, has been widely documented (Omasombo Tshonda, 2018).

Uranium-bearing phases are known to occur within several Cu-Co mineralized systems of the Copperbelt. Their presence contributes to the natural terrestrial radiation recorded in mining environments and may increase exposure to ionizing radiation for artisanal miners and nearby

communities. This issue is especially relevant at Luisha, where residential areas are located close to the open-pit workings.

The study therefore focuses on determining whether the available radiometric measurements are robust enough to support further analysis. Rather than relying on a single diagnostic, the assessment integrates statistical testing, geostatistical uncertainty analysis, signal-to-noise improvement, spatial visualization and mapping. This sequence is intended to identify weaknesses in the dataset before any modelling stage and thereby reduce the likelihood of misleading interpretations.

II. MATERIALS AND METHODS

Field acquisition combined a Geiger counter with GPS positioning along ten profiles selected with reference to the local geology. Prior to the survey, the ambient background level was measured at the Department of Geology, University of Lubumbashi, giving $0.41 \mu\text{Sv/h}$. Count-rate observations expressed in counts per minute were converted to $\mu\text{Sv/h}$ using the instrument conversion coefficient of 0.00833. Reconnaissance of the site was carried out beforehand to define accessible routes and profile orientations.

The analytical workflow comprised two complementary components. First, geostatistical diagnostics were used to examine prediction uncertainty, the frequency distribution of the observations and their spatial dispersion. Second, frequency-domain processing based on the Fast Fourier Transform was applied to suppress unwanted high-frequency fluctuations and improve the signal-to-noise ratio. OriginLab was used for spectral filtering, whereas ArcGIS supported geostatistical analysis and map production. The resulting outputs were then examined jointly to assess the consistency and reliability of the Luisha measurements.

III. GEOGRAPHICAL SETTING OF THE STUDY AREA

The investigated area occupies part of the Luisha mining district in Kambove Territory, Haut-Katanga, approximately

80 km northwest of Lubumbashi. Its geographic limits extend from about $27^{\circ}00'35''$ to $27^{\circ}02'21''$ E and from $11^{\circ}09'04''$ to $11^{\circ}11'09''$ S (Fig. 1).

Climatically, the area experiences a dry tropical regime with a marked seasonal contrast. Rainfall is concentrated between October and mid-April, whereas the dry season extends from mid-April to September. Annual precipitation is generally between 1,000 and 1,200 mm, with an average annual temperature close to 21°C and a mean atmospheric pressure of about 878 mbar (Malaisse, 1994).

The natural vegetation consists mainly of wooded savanna and open woodland, with trees locally reaching 15–20 m. Cu- and Co-rich substrates are also associated with distinctive low herbaceous vegetation commonly referred to as copper grassland or poisoning clearings. Highland steppe, riparian evergreen galleries and dense tall-grass savannas known locally as Dembo are also present.

Surface drainage reflects both relief and geological structure. Kabange places Luisha on a northwest-trending watershed separating tributaries of the Lualaba and Lupoto systems. Within the study area, drainage includes the Koroano and Luafi streams, both tributaries of the Lupoto.

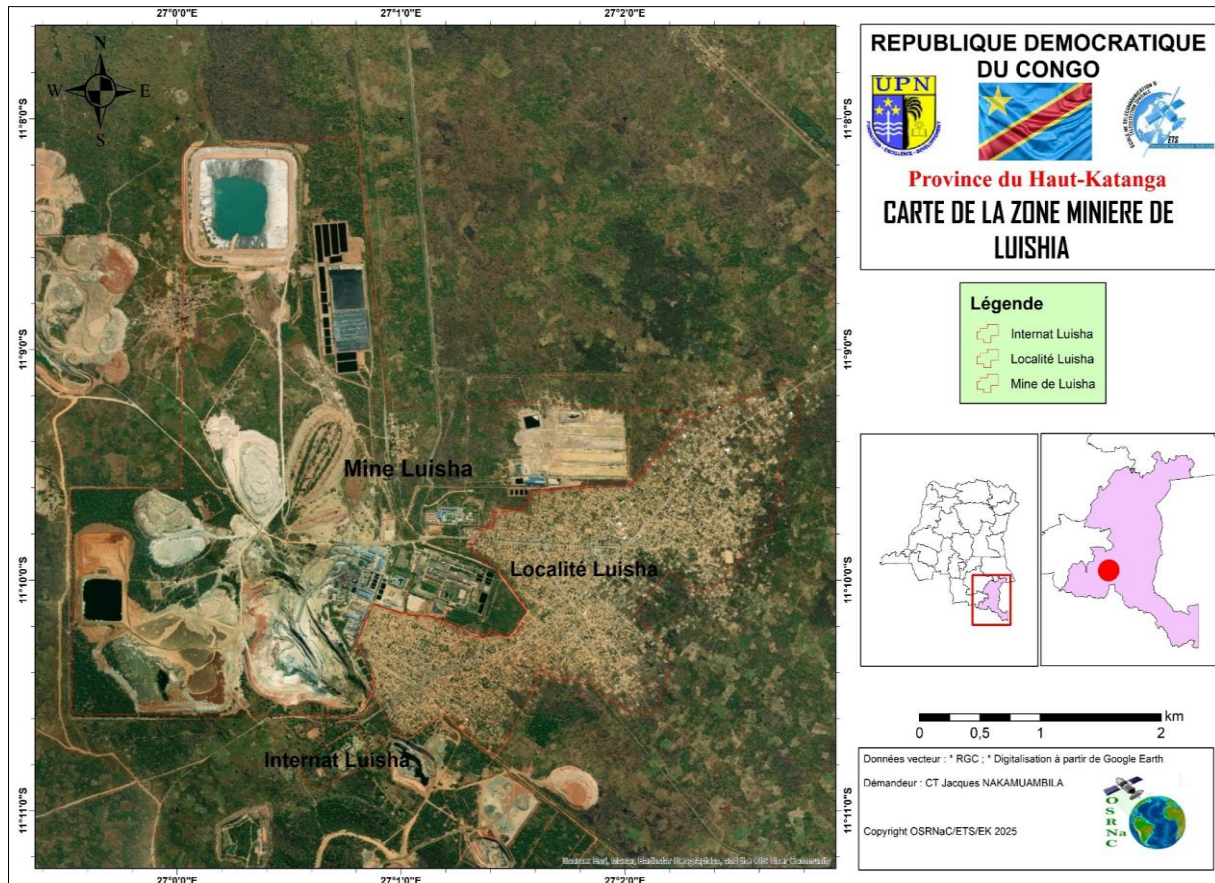


Fig 1 Location of the Study Area

IV. SPATIAL DISTRIBUTION OF THE RADIOMETRIC PROFILES

The survey network was organized into ten named radiometric profiles: AN, CM, CMI, CP, CT, HT, KZ, LB, UZ and WG. Each profile was reconstructed by joining successive measurement stations, allowing its position and geometry to be visualized across the mining area (Fig. 2).

Taken together, the profiles sample several parts of the site rather than a single corridor. Their orientations include east-west, north-south and oblique directions, which increases the probability of intersecting contrasting geological and surface conditions and improves the spatial representativeness of the survey.

The geometry of the survey lines is variable. CMI, HT and WG follow relatively regular trajectories, whereas UZ, CT and LB are more segmented or sinuous. Representing these profiles spatially is therefore important for linking individual measurements to their field position and for interpreting differences observed between radiometric curves.

Results were examined at profile scale by combining station measurements with radiation-versus-distance plots. To avoid overloading the paper, the full set of profiles is not reproduced in detail; instead, CMI, which contains 18 stations, is used as a representative example (Table 1).

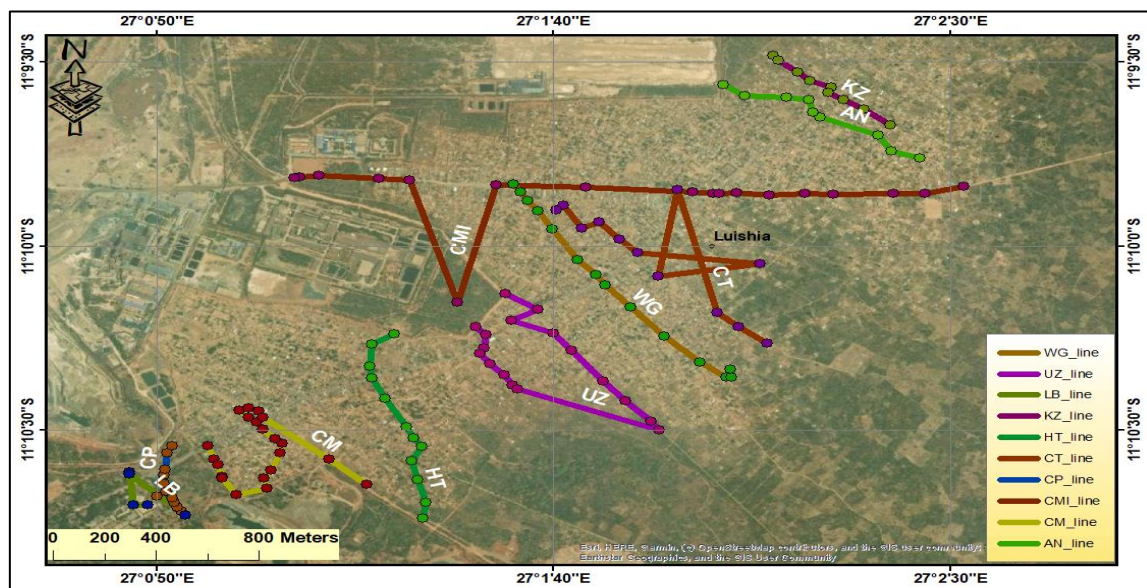


Fig 2 Spatial Distribution of the Radiometric Survey Profiles

Table 1 Geographic Coordinates, Lithology and Natural Terrestrial Radioactivity along Profile CMI.

Station	Longitude (°)	Latitude (°)	Altitude (m)	Lithological description	Natural radioactivity (cpm)	Natural radioactivity (μSv/day)
CMI001	27.01888	-11.16355	1237	Dolomite	84	2.8
CMI002	27.01872	-11.16357	1251	CMN debris	54	0.4
CMI003	27.01958	-11.16349	1250	CMN debris	81	2.6
CMI004	27.02167	-11.16362	1252	CMN debris	52	0.2
CMI005	27.02273	-11.16368	1253	CMN debris	69	1.6
CMI006	27.02441	-11.16923	1254	CMN debris	87	3.1
CMI007	27.02578	-11.16389	1254	CMN debris	153	8.6
CMI008	27.0289	-11.16402	1256	CMN debris	67	1.4
CMI009	27.03267	-11.16422	1258	CMN debris	62	1.0
CMI010	27.03336	-11.16428	1258	CMN debris	88	3.2
CMI011	27.03359	-11.16429	1258	Dolomite	99	4.1
CMI012	27.03421	-11.16427	1258	Dolomite	71	1.8
CMI013	27.03532	-11.16437	1259	Shale	83	2.8
CMI014	27.03658	-11.16429	1260	Shale	83	2.8
CMI015	27.03760	-11.16433	1259	Shale	74	2.0
CMI016	27.03968	-11.16429	1261	Shale	92	3.5
CMI017	27.04079	-11.16428	1259	Shale	78	2.4
CMI018	27.04214	-11.16399	1257	Shale	77	2.3

Background Value Reported for the Luisha Dataset: $BF = 0.41$.

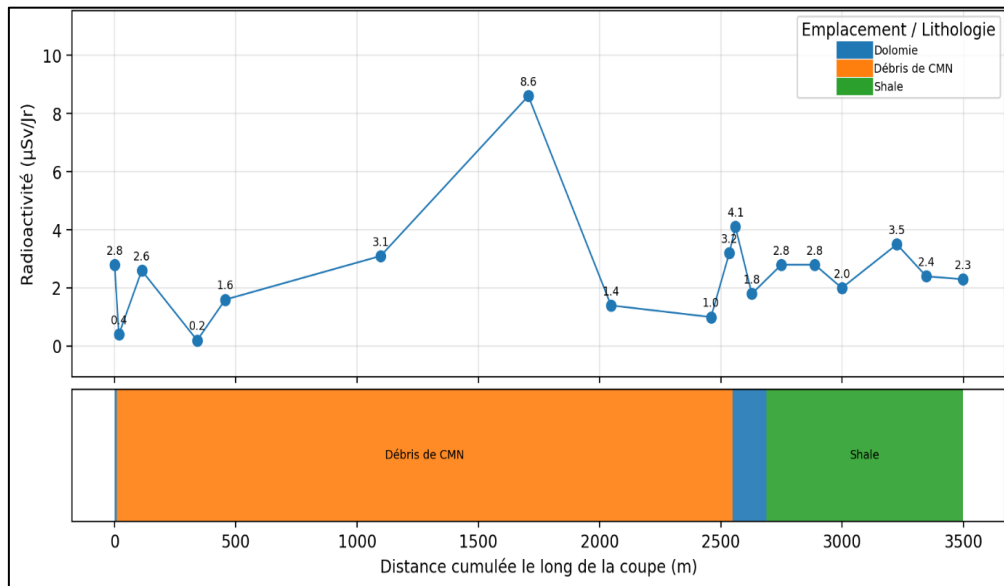


Fig 3 Variation in Natural Terrestrial Radioactivity ($\mu\text{Sv/day}$) with Distance and the Corresponding Lithological Band along Profile CMI.

CMI is the longest survey line, extending for approximately 3,498 m. It passes near the wastewater settling basin of the open-pit mine and crosses three main surface materials: CMN debris, described as brown sandy soil containing mine fragments and Black-Mineral Limestone; dolomitic material associated with reddish clayey soil; and shale developed beneath reddish-brown clayey soil. Along this profile, natural terrestrial radioactivity ranges from 0.2 $\mu\text{Sv/day}$ at CMI004 to 8.6 $\mu\text{Sv/day}$ at CMI007, with an average of 2.58 $\mu\text{Sv/day}$. This corresponds to an annualized dose of about 0.8 mSv/year, a value that warrants attention when compared with the 1 mSv/year reference level cited from the IAEA.

V. RESULTS AND ASSESSMENT OF RADIOMETRIC DATA RELIABILITY

➤ Geostatistical Analysis

Geostatistics was used here primarily as a data-quality tool rather than only as an interpolation technique. The ArcGIS Geostatistical Analyst environment was employed to examine how prediction confidence changes across the survey area and to identify locations where the available measurements provide stronger or weaker spatial support.

Kriging was selected because it estimates values at unsampled locations while providing an explicit measure of prediction uncertainty (Gratton, 2003). Mapping this uncertainty makes it possible to evaluate where the interpolated radiometric surface is strongly constrained by nearby observations and where it is more speculative (Fig. 4).

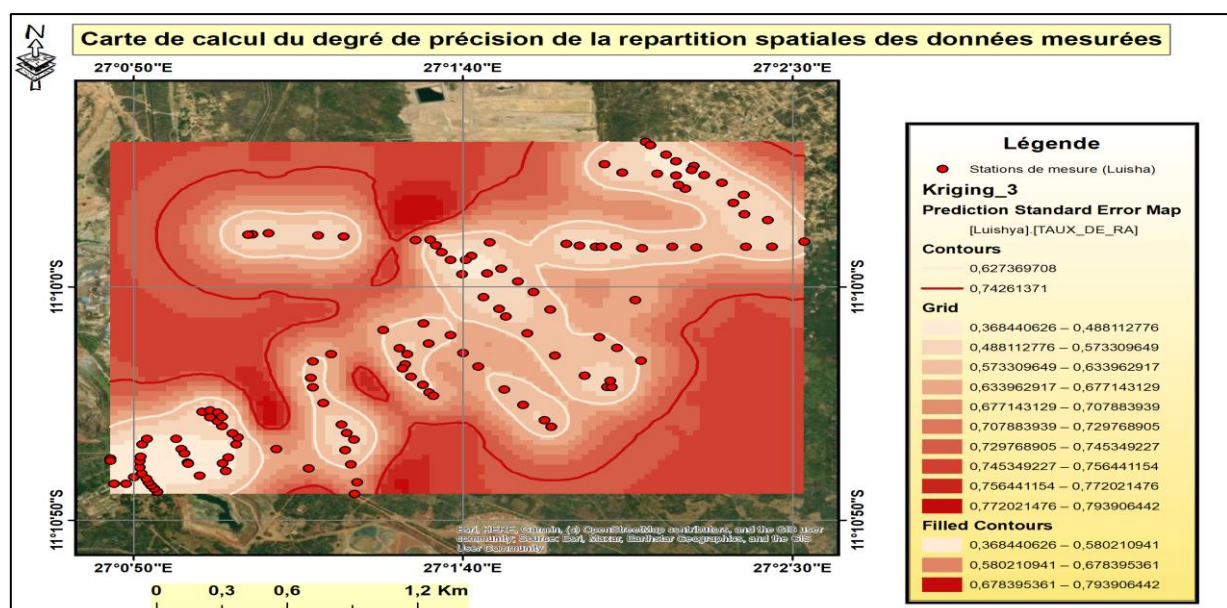


Fig 4 Standard Prediction Error Map for the Radiometric Dataset

Fig 4 should therefore be read as an uncertainty map, not as a radioactivity map. Areas shown in lighter tones have the lowest kriging standard errors, approximately 0.37-0.57, and generally coincide with sectors where measurements are relatively dense or closely spaced. Darker red zones reach about 0.73-0.79 and occur mainly farther from observations or between separated clusters of stations, where interpolation is less tightly controlled.

The spatial pattern of prediction error provides a practical criterion for judging map reliability and for locating areas where additional field measurements would be most

useful. In this representation, red indicates greater prediction uncertainty; it must not be interpreted as evidence of higher radiation levels.

The frequency distribution provides a complementary view of data quality. The Luishya histogram (Fig. 5) departs from perfect normality: most observations occupy the lower part of the range, while a relatively sparse upper tail extends toward higher readings. This right-skewed pattern indicates that a limited number of stations carry unusually large radiometric values.

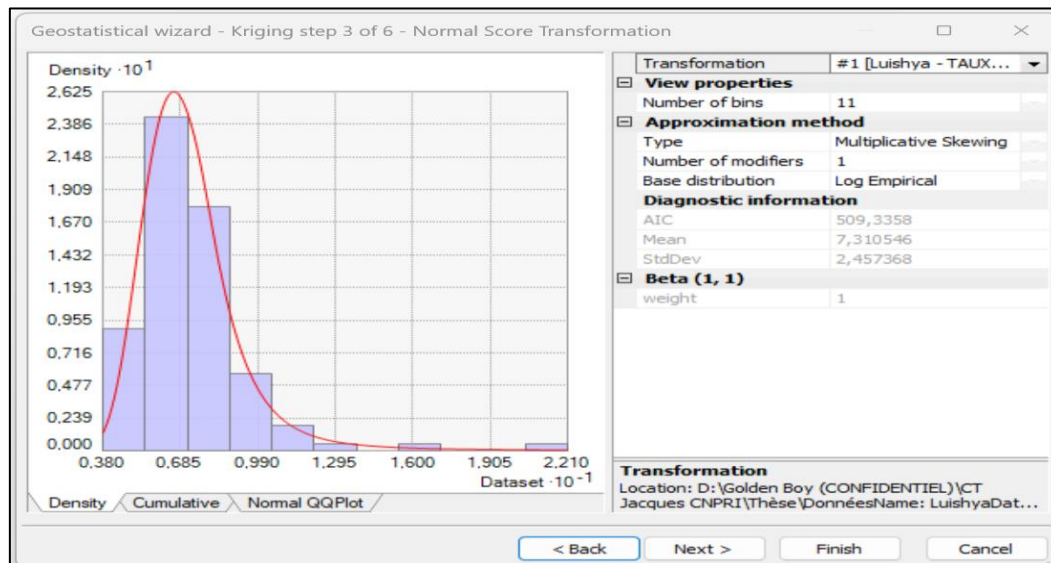


Fig 5 Histogram of the Radiometric Measurements

Most of the histogram density lies between approximately 0.5 and 0.9 in the units used on the original plot. Frequencies decrease rapidly beyond roughly 1.0-1.3, and only isolated observations approach 2.210. Such a distribution is more consistent with a positively skewed or near-lognormal population than with a strictly Gaussian one. Robust thresholds such as the 90th and 95th percentiles, or a logarithmic transformation where appropriate, are therefore preferable when the aim is to identify anomalously high values.

Normality was further examined with a Q-Q plot generated during the transformation stage of the ArcGIS kriging workflow (Fig. 6). This diagnostic compares the empirical quantiles of the radiometric dataset with those expected under a normal distribution and therefore highlights departures from Gaussian behaviour, particularly at the tails.

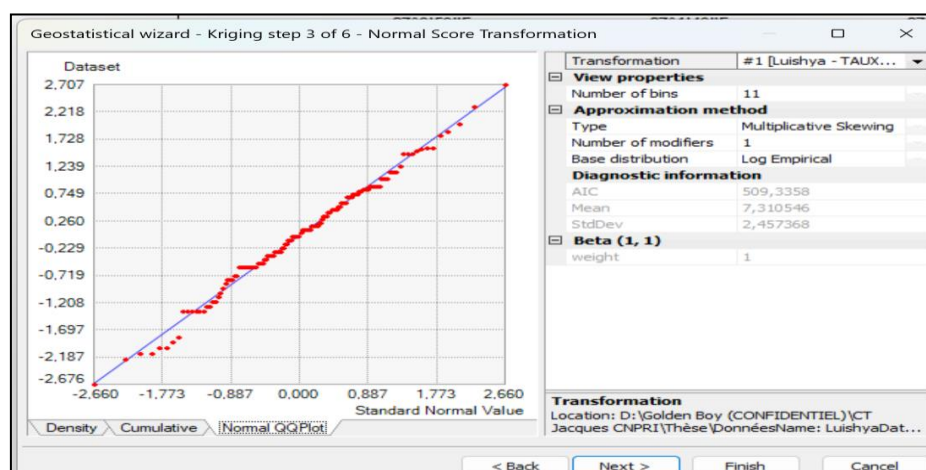


Fig 6 Normal Q-Q Plot of the Radiometric Data

The central portion of the Q-Q plot follows the theoretical line reasonably closely. This suggests that the bulk of the observations behaves approximately normally. The displayed diagnostic values are a mean of about 7.31 $\mu\text{Sv/h}$ and a standard deviation of about 2.46 $\mu\text{Sv/h}$. Noticeable departures occur at both ends of the plot, confirming that the extreme low and high observations are not fully described by a normal model.

For the geostatistical modelling stage, ArcGIS therefore used a Normal Score Transformation based on a Multiplicative Skewing approximation and a Log Empirical base distribution. This transformation reduces the influence

of the skewed tails and makes the data more compatible with the distributional assumptions used in kriging, without implying that the original measurements themselves are perfectly normal.

Sampling density was also mapped independently (Fig. 7). Low-density sectors are shown in green, whereas progressively warmer colours indicate increasing concentrations of measurement stations. This map provides a direct visual link between field sampling geometry and the confidence that can reasonably be assigned to interpolated results.

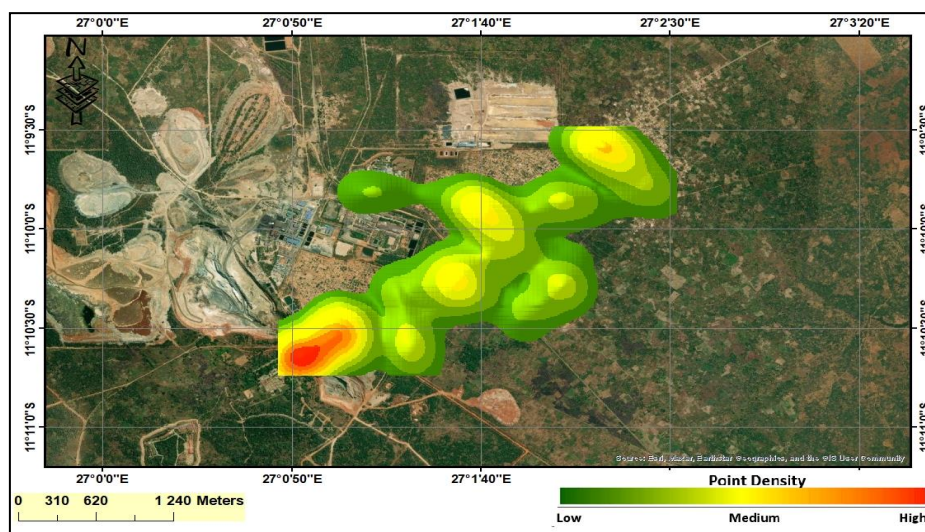


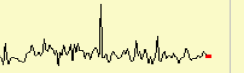
Fig 7 Density of Radiometric Measurement Stations at the Luisha Site (ArcGIS).

The densest cluster occurs in the southwestern sector of the site, with additional zones of moderate station concentration in the central, north-central and northeastern parts. Because the network is uneven, interpolation confidence is likewise spatially variable. Areas with dense sampling are better constrained, whereas sparsely measured sectors would benefit most from targeted infill observations.

➤ Signal-to-Noise Ratio Enhancement

A low-pass FFT procedure was applied to reduce high-frequency noise in the Luisha radiometric series. The filtering configuration used a two-point window and a cutoff frequency of 0.25 Hz. The purpose was not to alter the broad radiometric trend, but to suppress rapid irregular fluctuations that could obscure meaningful variations in the signal (Table 2).

Table 2 Results of Low-Pass FFT Filtering Applied to the Luisha Radiometric Dataset for Signal-to-Noise Enhancement

Results of Data Post-Processing Applied to the National Radiometric Database for Signal to Noise Estimation							
Long Name	A(X)	B(Y)	C(Y)	D(Y)	E(Y)	G(Y)	H(Y)
N°	STATION WAY-POINT	LATITUDE	LONGITUDE	ALTITUDE	TAUX DE RADIOMETRIE	TAUX DE RADIOMETRIE (Filtré)	
Units		(°)	(°)	(m)	($\mu\text{Sv/Jr}$)		
Comments						2 pts Low-Pass FFT smooth of "TAUX DE RADIOMETRIE"	
Sparklines							
1	1 LB001	-11.17838	27.01357	1236	4.4	5.78752	
2	2 LB002	-11.17838	27.01306	1225	7	5.64103	
3	3 LB003	-11.17688	27.01292	1241	4.4	5.64284	
4	4 LB004	-11.17701	27.01292	1245	6.2	5.6504	
5	5 LB005	-11.17885	27.01488	1255	5.8	6.06571	
6	6 CP001	-11.17866	27.01475	1257	7.6	7.32289	
7	7 CP002	-11.17848	27.01462	1259	5.6	8.95701	
8	8 CP003	-11.17829	27.01452	1260	15.6	9.70871	
9	9 CP004	-11.17808	27.01443	1261	7.4	8.88023	
10	10 CP005	-11.17798	27.0139	1261	6.5	7.26413	
11	11 CP006	-11.17777	27.01422	1262	6.1	6.29651	
12	12 CP007	-11.17741	27.01415	1262	4.2	6.43967	
13	13 CP008	-11.17703	27.01415	1263	10.5	6.85994	
14	14 CP009	-11.17677	27.01418	1265	6.2	6.68309	
15	15 CP010	-11.17604	27.01426	1265	4.3	6.05029	
16	16 CP011	-11.17571	27.01445	1266	6.1	5.70289	
17	17 CM001	-11.17571	27.01567	1266	6.4	5.91511	
18	18 CM002	-11.17632	27.01589	1275	6.2	6.24484	

The filtered series is visibly smoother than the raw measurements and therefore easier to interpret. Increasing the smoothing window generally strengthens attenuation of short-wavelength components, while lowering the cutoff frequency removes a larger fraction of the high-frequency content. In this dataset, the selected settings provided a compromise between noise suppression and preservation of the overall radiometric pattern.

After FFT filtering, cross-validation was used as an independent check on the geostatistical model (Fig. 8). The aim was to determine whether predictions remain consistent with measured values once the spatial model and preprocessing steps are taken into account.

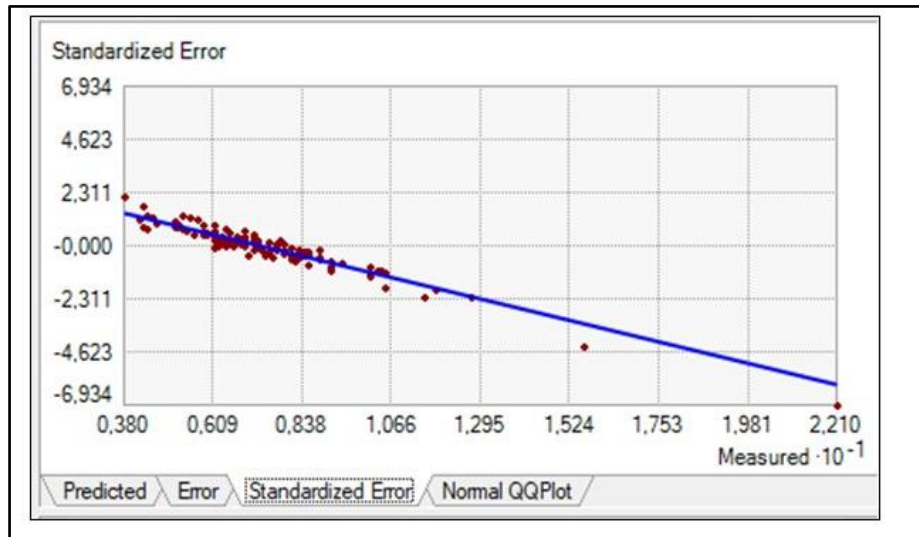


Fig 8 Cross-Validation Plot for the Luisha Radiometric Measurements

For most stations, standardized errors remain relatively small and the observations lie close to the fitted trend, indicating acceptable agreement between measurements and predictions. A few high-value observations deviate more strongly, showing that uncertainty increases around the most extreme readings. Consequently, the model performs satisfactorily for the bulk of the dataset, while the strongest anomalies should still be interpreted cautiously.

VI. DISCUSSION

Taken together, the quality-control tests show that no single metric is sufficient to characterize the Luisha dataset. Reliability reflects the interaction of sampling geometry, distributional behaviour and signal noise. The density and prediction-error maps demonstrate that spatial confidence is greatest where stations are numerous and closely spaced, whereas uncertainty increases in poorly sampled sectors. This relationship confirms that part of the interpolation error is controlled by survey design rather than by the radiometric variable alone.

The statistical diagnostics point to a second limitation: the measurements are only approximately normal. The central population is well behaved, but positive skewness and tail departures are generated by a small number of high readings. These values may represent genuine local radiometric anomalies associated with lithology, mine-related materials or near-surface alteration; alternatively, they may exert disproportionate influence on conventional statistical measures. Their geological context should

therefore be considered alongside their statistical classification.

Signal processing provides a third line of evidence. The low-pass FFT filter reduced rapid high-frequency oscillations without removing the broader spatial trend, and the subsequent cross-validation remained satisfactory for most observations. The combined evidence supports use of the dataset for further interpretation, but with two qualifications: prediction in sparsely sampled zones carries greater uncertainty, and the largest radiometric anomalies require separate scrutiny rather than being treated as ordinary members of the background population.

VII. CONCLUSION

Radiometric data from 128 stations distributed among ten profiles at the Luisha mining site were subjected to a multi-step quality assessment before further interpretation. The workflow brought together spatial sampling analysis, descriptive statistics, geostatistical uncertainty modelling and frequency-domain filtering, allowing weaknesses in both the survey geometry and the measured signal to be identified.

The results indicate that spatial prediction quality is closely tied to station density. Statistical plots reveal a largely regular central population accompanied by a small number of extreme values, while kriging identifies where interpolation uncertainty is lowest and highest across the site. Low-pass FFT processing reduces high-frequency noise and improves readability of the radiometric signal. On this basis, the dataset is sufficiently robust for subsequent mapping and radiometric

modelling, provided that poorly sampled sectors and the strongest anomalies are treated with appropriate caution in any assessment of natural ionizing-radiation exposure.

- [13]. OriginLab Corporation. FFT Filter / Signal Processing Algorithms – Origin Documentation.

REFERENCES

- [1]. Gratton, Y. (2003). Le krigeage : la méthode optimale d'interpolation spatiale. INRS-Eau-Terre-Environnement, Québec, Canada.
- [2]. Omasombo Tshonda, J. (2018). Haut-Katanga : Lorsque richesses économiques et pouvoirs politiques forcent une identité régionale. Tome 2 : Bassin du cuivre : matrice et horizon. MRAC, Tervuren.
- [3]. Kabange (1982). [Reference details as cited in the source manuscript].
- [4]. UNSCEAR (2000). Sources and Effects of Ionizing Radiation – Annex B: Exposures from Natural Radiation Sources. United Nations Scientific Committee on the Effects of Atomic Radiation, New York.
- [5]. IAEA (1998). Characterization of Radioactively Contaminated Sites for Remediation Purposes. IAEA-TECDOC-1017, International Atomic Energy Agency, Vienna.
- [6]. IAEA (2004). Remediation of Sites with Dispersed Radioactive Contamination. Technical Reports Series No. 424, International Atomic Energy Agency, Vienna.
- [7]. Yeşilkanat, C. M., Kobya, Y., Taşkın, H., & Çevik, U. (2015). “Dose rate estimates and spatial interpolation maps of outdoor gamma radiation with geostatistical methods; a case study from Artvin, Turkey.” *Journal of Environmental Radioactivity*.
- [8]. Sanusi, M. S. M., Ramli, A. T., Gabdo, H. T., et al. (2014). “Isodose mapping of terrestrial gamma radiation dose rate of Selangor State, Kuala Lumpur and Putrajaya, Malaysia.” *Journal of Environmental Radioactivity*.
- [9]. Le Coz, M., et al. (2021). “Factorial kriging for estimating and mapping the geochemical background component of gamma dose rate around former uranium mines.” *Journal of Environmental Radioactivity*.
- [10]. Borgoni, R., De Francesco, D., De Bartolo, D., Tzavidis, N., & Giannardi, C. (2011). “A geostatistical approach to assess the spatial association between indoor radon concentration, geological features and building characteristics: the case of Lombardy, Northern Italy.” *International Journal of Environmental Research and Public Health*, 8(5), 1420–1440.
- [11]. Dinis, M. L., et al. (2021). “Assessment of natural radioactivity, heavy metals and radiation exposure at a former uranium mining site.” *Atmosphere*, 12(11), 1433.
- [12]. Expósito-Suárez, V. M., et al. (2024). “Radiological characterization of the tailings of an abandoned copper mine with high levels of naturally occurring radionuclides.” *Environmental Geochemistry and Health*.